Devin Corti

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Technical Skills

Mechanical: SolidWorks, AutoCAD, Fusion 360, Inventor, Welding, Robots, Machining, 3D Printing, Laser Cutting
Firmware: C++, Beckhoff TwinCAT, ABB Rapid, PLC Ladder, MATLAB, Git, Python, JavaScript, Excel
Electrical: PLC, IO Link, Robotic Systems, Wiring, Electrical Components, Circuits, Arduino
Certifications: WHMIS 2015, Class 5 Driver's Licence (AB), DELF B2 (French), Worker Health & Safety Awareness

Experience

Failure Analysis Intern — Sept. 2026 – Pres.
Tesla Inc. — *Palo Alto, CA*

- Designing in-house solutions to streamline existing failure analysis workflows
- Performing functional testing and board level debug to reproduce symptoms and confirm electrical failure modes
- Determine root cause for PCBA/Component level failures by leveraging failure analysis lab tools and techniques
- Performing hands on DOE to confirm root causes and corrective actions and creating written reports that clearly communicate results of failure analysis.

Research and Development Engineering Co-op — Jan. 2026 – May 2026
Magna International — *Brampton, ON*

- Designed and built a machine vision camera system aimed at improving defect detection in automotive parts.
- Improved machine vision model performance by optimizing lighting, model structure, and training methods.
- Participated in the development of a robotic weld inspection system to improve part quality and cycle time.
- Programmed a prototype RFID inventory management system using a PLC to track material around plants.
- Led experiments to determine the effects of coatings on the bonding of metals for automotive manufacturing.

Robotics Design Co-op — May 2025 – Aug. 2025
RoBIM Technologies Inc. — *Edmonton AB.*

- Designed, built, and tested a multifunctional tool splitter using Fusion360 that combined a vacuum gripper and stapler end effector, streamlining robotic tool changes and reducing cycle time in wall panel manufacturing.
- Led the design of a robotic material preparation table, allowing automated cutting of plywood and lumber to expand robotic capabilities in prefabricated wall panel production
- Collaborated on PLC integration efforts, working to connect a PLC controller with an ABB industrial robot to expand sensor I/O and improve system control flexibility

Extracurriculars

University of Waterloo Formula Electric — Sept 2024 – Present
Pedals Lead — *Waterloo, ON*

- Leading the design and development of the pedal system, balancing reliability, manufacturability, and performance
- Designed and validated pedal box components using CAD and FEA, iterating designs based on structural and packaging requirements.

Seats Lead

- Designed a carbon fibre racing seat using surface modeling tools, and a mould for manufacturing.
- Built the seat using a layup method, decreasing the weight of the seat by 50%, and improving driver ergonomics.

Robotics Team Lead — Sept. 2022 – Feb. 2024
Springbank Robotics Team — *Calgary, AB*

- Led the team to the FIRST World Championships by designing, building and testing of the robot
- Created a CAD model of the robot using Fusion 360 to ensure the robot could be built without complication

Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science in Mechatronics Engineering — *Sept. 2024 – Apr. 2029*

- **Overall GPA:** 98%
- **Scholarships & Awards:** First In Class, Carl A. Pollock Scholarship, President's Scholarship of Distinction.

Other awards: Valedictorian, Schulich Leader Nominee, Governor General's Bronze Medal, Alberta Centennial Award
Other Involvement: Varsity Rugby, Academic Rep, Hockey, Baseball, Refereeing
Interests: NHL (Habs), MLB (Jays), Travel, Reading, Hiking, Cooking, Formula 1, Chess